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Symbolic Logic Framework for Situational Awareness in Mixed Autonomy

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D7.1 Plan for Dissemination and Exploitation

Work Package: WP7

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Contents

1 Introduction....3
2 Dissemination strategy and approach....4
3 Completed and on-going dissemination activities....7
3.1 Scientific publications and software tools....7
3.2 Social media visibility and project website....7
3.3 SymAware community building events....8
3.4 Portfolio activities....9
3.5 Advisory board....9
4 Future dissemination activities....10
5 Exploitation strategy and approach....12
5.1 Introduction....12
5.2 Building network of SymAware Followers....12
5.3 Software releases and workshops....13
5.4 Exploitation plans from SymAware partners....13
6 Completed and on-going exploitation activities....14
6.1 Building network of SymAware Followers....14
6.2 Software releases and workshops....14
6.3 Exploitation plans from SymAware partners....14
7 Future exploitation activities....15
7.1 Building network of SymAware Followers....15
7.2 Software releases and workshops....15
7.3 Exploitation plans from SymAware partners....15
Appendix....16
A.1 Scientific publications and software tools....16
A.2 SymAware community building events....19

1 Introduction

This document serves as a first effort to provide a comprehensive dissemination and exploitation strategy for the SymAware project. This is a "living" document which will be updated at M12 and M36 to include the completed, ongoing, and planned results of the dissemination and exploitation activities, the evaluation of the dissemination and exploitation strategies and the actions taken to improve it. This allows us to adapt our activities based on the project progress, intermediate results and opportunities that arise on the outside.

The deliverable consists of two main parts: plan for dissemination activities and plan for exploitation activities. The contents of the deliverable have been divided into 7 sections:

1. Introduction: A brief description of the deliverable, as well as the purpose of the document. logical characterization of awareness using symbolic methods,

2. Dissemination strategy and approach: An overview of the strategy to be followed to disseminate the results and actions of the SymAware project; A description on the initial dissemination plan, describing the key messages and scientific areas, the target audience of the SymAware dissemination strategy, the channels, and the expected results of the plan.

3. Completed and on-going dissemination activities: A detailed list of the completed and on-going dissemination activities and developments following the dissemination strategies.

4. Future dissemination activities: A plan for new activities to be conducted for future developments during the life of the project.

5. Completed and on-going exploitation activities.

6. Future exploitation activities.

2 Dissemination strategy and approach

Dissemination activities are key in SymAware and they were managed by asking the collaboration of all partners. This demonstrates a strong commitment to raise awareness of the project, to different target audiences. Efficient dissemination is fundamental in any research process, as the success of projects outcomes is largely connected to dissemination activities. Careful and early planning of dissemination activities therefore is very important.

A dissemination strategy was created to be carried out along with the main scientific outputs of the project. The dissemination plan breaks down 36 months of the project into 3 reports that categorize the strategic activities completed. For each dissemination report, events are classified into appearances, publications, and promotional material within the stated 6 months. Information regarding the distributor, date, event, activity, the targeted group and attendee are provided in chronological order. Additional information with the link and a short description of the event is included in the chart for quick reference. In addition, a dissemination plan for the next 12 months is provided at the end of the document.

The dissemination of SymAware is based on achievement of work packages W1 to W8 from months 1 to 36 of the project. These work packages are:

WP1: Situational Awareness
WP2: Spatiotemporal goal awareness
WP3: Risk Awareness
WP4: Knowledge awareness
WP5: Use cases
WP6: Requirements for trustworthy and ethical human-agent interaction
WP7: Exploitation, Communication and Dissemination
WP8: Project Management

Dissemination activities will be achieved and managed as part of WP7. Activities in this work package demonstrate the commitment of the consortium group in raising awareness of the project to the target markets. The main objectives of this WP are:

- Reporting and archiving results
- Constructing and updating the project website
- Promoting scientific dissemination
- Supporting intellectual property issues and commercialization

The deliverables for exploitation and dissemination under WP7 includes:

- D7.1 (M6) “Plan for Dissemination and Exploitation (first version)”
- D7.2 (M12) “Plan for Dissemination and Exploitation (second version)”
- D7.3 (M36) “Plan for Dissemination and Exploitation (final version)”
- D7.4 (M6) “Data Management Plan (first version)”.
- D7.5 (M36) “Data Management Plan (final version)”.

- D7.6 (M3) “Public website”, Design and first implementation with core (static) content.
- D7.7 (M12) “Software tools on public Git repository (first release)”.
- D7.8 (M24) “Software tools on public Git repository (second release)”.
- D7.9 (M34) “Software tools on public Git repository (third release)”.

Making the project and its deliverables attractive to stakeholders is a key part of SymAware and it is important to create clear activities that help to spread community awareness of the scientific and technical aspects of the project. The dissemination strategy for SymAware will include participation in events, stakeholder liaison, online activity, media and scientific dissemination. Different communication channels will be used in order to address all target groups. Efficient dissemination is a fundamental activity in any research process, as the success of dissemination activities contributes to project outcomes by all interested target groups.

Dissemination activities will be managed as part of WP7, where all project partners will be asked to contribute. The project consortium demonstrated commitment in this sense to raise awareness of the project. The following important actions were taken to maximize the impact of dissemination:

- identify the relevant target groups
- share the SymAware project idea in the most suitable way according to the needs of target groups
- identify any barriers in order to be able to make sure dissemination goes beyond these
- deliver dissemination materials and activities

Dissemination strategy

During the project, efforts will be made to disseminate project goals and results to a variety of groups, this includes the writing of scientific publications to communicate the results to the scientific community. Conferences and seminars will be planned and attended to target researchers, and other key audiences in the scientific communities. Several measures will be taken in order to disseminate results to target groups. Scientific publications will be written and sent to journals in order to communicate the results to the scientific community. Consortium members will be attending events and seminars to target stakeholder groups in the scientific community.

Papers, presentations and other dissemination activities

Presentations and papers were prepared by consortium partners that attended specific conferences and events. In particular:

- Workshops at…
- Conference papers presented at…
- Scientific papers…
- Other activities (including keynote speeches, exhibitions and conferences)

Further reports and scientific papers are to be published throughout each stage of the project, up until the project end, at M36.

SymAware Dissemination strategy

Information to disseminate

- SymAware goals and potential exploitable results

Dissemination

1. Scientific publications and software tools
   - Publications
   - Technical papers
   - Presentations
   - Reports
   - White papers
   - Development of software tools
2. Social media visibility and project website
- Online activity
  - Blog posts about ongoing activities and results
  - Mention of activities on social media channels
  - Email newsletters
  - Project website development
- Media
  - Press releases about the project, updates or milestones
  - Interviews
3. SymAware community building events
- Participation in events (exhibitions, meetings)
  - Explanation of the project
  - Presentations, keynote speeches, etc
  - Distribution of leaflets
- Liaison with stakeholders
  - Visits
  - Email updates
4. Portfolio activities
5. Advisory board

3 Completed and on-going dissemination activities

3.1 Scientific publications and software tools

The consortium contributed to a variety of scientific papers, publications and articles published during the period of the SymAware project. This included papers aimed at various different targets, such as the control community and robotics community.

See all of the Scientific Papers, Publications and articles published in Appendix A.1.

3.2 Social media visibility and project website

Promotional material distributed to support the dissemination of the SymAware project by consortium members included blogs, videos, websites and social media pages to mark various project milestones.

Project logo

An important aspect of establishing the project's identity was designing a project logo, which would be essential for use in dissemination material. A number of logos were designed and a project logo was agreed on by the consortium as follows.

Project website

The project website is a dissemination tool where key information about the project can be shared, for example details on members of the consortium, project objectives and dissemination activities such as publication of research papers. This provides key information for partners writing about or disseminating information about the project, a well as providing a source of information for external parties interested in finding out more about the project. The address for the website is https://www.symaware.eu/

The website will be updated regularly as the project progresses.

Social media tools

SymAware social media strategy aimed at providing stakeholders with information and new developments in the SymAware project. The aim of the posts is also to reach a wider audience. The social media platforms the project will use are LinkedIn, Twitter, and YouTube. SymAware has its own channels for each.

LinkedIn

SymAware has a LinkedIn account. All consortium members follow the SymAware LinkedIn page at: https://www.linkedin.com/company/symaware/ and share updates on their own LinkedIn accounts.

Twitter

All consortium members follow the SymAware Twitter page at: https://twitter.com/SymAware and repost messages on their own Twitter accounts. Sharing messages on the Twitter channel will involved sharing videos from the YouTube channel.

YouTube

SymAware has a YouTube channel at: https://www.youtube.com/@SymAwareProject/featured where relevant videos will been shared, including all project milestone videos, and project interviews and introductory videos.

3.3 SymAware community building events

The SymAware consortium led and or participated in a range of events throughout the project period which helped to disseminate the goals and outcomes of the SymAware project. This included annual events, talks, workshops and demonstrations in Europe, the US and Asia, between 2022 and 2025. Annual events with a large participation from the scientific community attended by consortium members included:

At some of these events, SymAware was focused on particularly, for example in a presentation of project highlights and in related talks and presentations.

Events consortium members attended during M1-M6 included:

- SymAware Kickoff Meeting
- ACM/IEEE International Conference on Human-Robot Interaction (HRI)
- CPS Workshop
- 61st IEEE Conference on Decision and Control

Find more information about events attended in Appendix A.2.

3.4 Portfolio activities

- Portfolio kick-off meeting on 7th and 8th December 2022 at Saint-Josse-ten-Noode Brussels, Belgium
- Meeting with another project partner in the portfolio (EIC Pathfinder Astound project) on possible collaboration and moving towards having a start up
- Steering committee meeting with portfolio members at the beginning of April 2023.

3.5 Advisory board

- Meetings will be arranged with the advisory board members to get the feedback on the progress of the project.
- MPI-SWS had a meeting with the independent ethical adviser (EA) on 16th March 2023
- UU had a meeting with the independent ethical adviser (EA) on 16th March 2023.

4 Future dissemination activities

Partner	Exhibitions, Demonstrations, Forums, Seminars etc.	Papers, publications and articles	Promotional material	Media engagement	Other
Report M07-M18					
MPI-SWS	Dr Sadegh Soudjanin is organizing an invited session at the CDC 2023 conference on resilience, safety, and security of CPS				
	Dr Sadegh Soudjanin is organising a workshop along with IFAC in Japan on the topic: Data-Driven Verification and Control of Cyber-Physical				
TU/e	Dr. Sofie Haesaert will be presenting at the Conference on Decision and Control 2023 on data driven control				
	Dr. Sofie Haesaert will be presenting				

	at the 21st European Control Conference (ECC) on data driven control				
KTH	Prof. Dr. Dimos Dimarogonas is organizing an invited session at the CDC 2023 conference on Formal Methods for Time-Critical Decision Making and Control				
UU					
NLR					
SISW					

5 Exploitation strategy and approach

5.1 Introduction

To maximize the chance of future uptake of project results, the consortium will empower key actors by building a SymAware community of followers, consisting of both academics specializing in related fields and especially young companies (start-up and scale-up phase) developing business in the field of autonomous systems. We will actively seek contact and stay in close communication with these stakeholders to (1) enable them to follow our developments, (2) provide them with releases of our developed software tools, (3) organize workshops with tutorials and examples during hands-on sessions using our developed software tools and (4) getting their feedback regarding the technology potential as they see it from their perspectives.

For building this community of followers we will disseminate the project details (goals, partners, structure) to a wide as possible target group of potentially interested parties (academics and industry) using many different communication channels. We will invite interested people to actively register as "SymAware follower".
This will result in a list of contact emails we will use for providing the community with:

- Updates on publications/presentation of SymAware researchers. In this way followers know at what Conference they could meet with SymAware researchers.
- Invitations to SymAware workshops. Workshops to work with SymAware developed software tools are scheduled in M12, M24 and M34.
- Newsletter to keep track of progress.

5.2 Building network of SymAware Followers

The target group of potential SymAware followers are academic groups in Computer Science and Control and industries with R&D staff profiles closely matching those of the SymAware researchers.
We will send out invitations to register as a SymAware follower using the following channels:

- All academic project partners using their own professional network.
- All industry project partners using their relevant customer contacts
- All portfolio partners partners which we will also ask to be ambassador of our project
- The invitation to register will be placed on our public website
- The invitation to register will be placed on the Portfolio website
- The invitation to register will be placed on our LinkedIn and Twitter channels
- The invitation to register will be placed on our project flyer that will distributed
- The invitation to register will be placed on all other dissemination and communications, where possible

5.3 Software releases and workshops

Key activity for increasing the changes for uptake of project results will be through actively guiding people in using the software tools that are based on the research outcomes of the project.

Three releases of the SymAware framework and tools integration have been scheduled in M12, M24 and M34. Soon after these release dates, workshops will be organized where SymAware followers can use the software framework under guidance of the SymAware researchers.

5.4 Exploitation plans from SymAware partners

Exploitation potential of project results for SymAware industrial partners will be identified and were appropriate further research directions could be taken such to further explore these exploitation potentials.

The same holds for researchers at academic partners that see potential to further develop project results into novel products or services through start-up companies.

6 Completed and on-going exploitation activities

6.1 Building network of SymAware Followers

The following activities are already on-going:

1. Setting up an online registration tool to collect follower registrations.
2. Setting up an online newsletter template to be able to send regular updates to the SymAware followers.
3. Creation of a project flyer to be posted on the social media channels and to be handed out during conferences/event

6.2 Software releases and workshops

6.3 Exploitation plans from SymAware partners

7 Future exploitation activities

7.1 Building network of SymAware Followers

- Preparation of first followers workshop at M12 (September 2023).

7.2 Software releases and workshops

- Prepare for the version 1 software release and Workshop in M12 (September 2023).

7.3 Exploitation plans from SymAware partners

No update

Appendix

A.1 Scientific publications and software tools

Partner	Title	Author	Year and Magazine	Target	Status	Joint publication
Report M01-M06						
MPI-SWS	Symbolic Knowledge Awareness for Autonomous Driving	Andrea Calvagna, Arabinda Ghosh, and Sadegh Soudjnai	2023	CDC 2023	Under preparation/ submission	No
	SySCoRe: Synthesis via Stochastic Coupling Relations	Birgit van Huijgevoort, Oliver Schön, Sadegh Soudjani, Sofie Haesaert	2023	HSCC 2023	Accepted	Yes
	Correct-by-Design Control of Parametric Stochastic Systems	Oliver Schön, Birgit van Huijgevoort, Sofie Haesaert, and Sadegh Soudjani	2022	CDC 2022	Accepted	Yes
	Data-Driven Abstraction-based Control Synthesis	Milad Kazemi, Rupak Majumdar, Mahmoud Salamati, Sadegh Soudjani, Ben Wooding	2023	Nonlinear Analysis Hybrid Systems	Under submission	Yes
	A Toolchain for Accelerated Symbolic Control	Rupak Majumdar, Kaushik Mallik,	2023	HSCC	Accepted as poster	Yes

	(Software tool)	Mateusz Rychlicki, Anne-Kathrin Schmuck, Sadegh Soudjani				
	A Toolchain for Accelerated Symbolic Control (Software tool paper)	Rupak Majumdar, Kaushik Mallik, Mateusz Rychlicki, Anne-Kathrin Schmuck, Sadegh Soudjani	2023	CAV	Under submission	Yes
TU/e	SySCoRe: Synthesis via Stochastic Coupling Relations	Birgit van Huijgevoort, Oliver Schön, Sadegh Soudjani, Sofie Haesaert	2023	HSCC 2023	Accepted	Yes
	Correct-by-Design Control of Parametric Stochastic Systems	Oliver Schön, Birgit van Huijgevoort, Sofie Haesaert, and Sadegh Soudjani	2022	CDC 2022	Accepted	Yes
KTH	Distributed control synthesis of signal temporal logic tasks for multi-agent systems	Siyuan Liu, Dimos Dimarogonas	2023	Control engineering community	Under preparation/ submission	No
	High-order control barrier functions for a prototype multi-spacecraft inspection of the ISS	Gregorio Marchesini, Pedro Roque, Dimos Dimarogonas	2023	CDC 2023	Under preparation/ submission	No
UU	A Case Study in Designing Trustworthy Interactions and its Implications for Socially	Mengyu Zhong, Marc Fraile, Ginevra Castellano and	2023	Frontiers	Under review	No

	Assistive Robotics	Katie Winkle				
	Trustoworthy driver-autonomous vehicle interaction: a participatory design study on the ethical considerations	Ana Tanevska, Ginevra Castellano and Katie Winkle	2023	HAI 2023	Under preparation	No
NLR						
SISW						

A.2 SymAware community building events

Partner	Event	Activity	Date & Location	Attendee
Report M01-M06				
MPI-SWS	Cyber-physical Systems Workshop 2023	Dr Sadegh Soudjani has given a talk on "Learning Approaches for Formal Synthesis of Cyber-Physical Systems" in line with the SwmAware project activities	March 21st 2023 at MPI-SWS, Kaiserslautern, Germany	Researchers and scientific members
	One-day workshop	Dr Sadegh Soudjani hosted Prof Raphael Jungers	2nd march 2023 at Newcastle University, UK	Researchers and scientific members
	Research collaboration	Dr Sadegh Soudjani visited the robotics Lab of Sheffiled University	January 4th and 5th 2023 at Sheffield, UK	Researchers and scientific members
	Presentation on learning methods for formal verification of CPS	Dr Sadegh Soudjani gave a presentation on learning methods for formal verification of CPS at Sheffield University	December 8th, 2022 at Sheffield, UK	Researchers and scientific members
	Presentation on learning methods for CPS	Dr Sadegh Soudjani gave an internal presentation on learning methods for CPS at MPI-SWS	November 8th, 2022 at MPI-SWS, Kaiserslautern, Germany	Researchers and scientific members
TU/e	Presentation on data-driven verification for the future of mobility	Dr.ir. Sofie Haesaert gives a presentation at the Women in Data Science event in Eindhoven	March 27, 2023, High-tech Campus Eindhoven	Researchers, Engineers

KTH	Presentation on formal methods based control synthesis of CPS	Prof. Dr. Dimos Dimarogonas and Dr. Siyuan Liu attended and gave presentation at the 61st IEEE Conference on Decision and Control	December 6-9, 2022, in Cancun, Mexico	Researchers and scientific members
	Research collaboration	Dr. Siyuan Liu visited the Control Systems Group of Eindhoven University of Technology	December 18-22, 2022, in Eindhoven, Netherlands	Collaborators, researchers and scientific members
UU	Organization of the 2023 ACM/IEEE International Conference on Human-Robot Interaction (HRI 2023)	Prof. Dr. Ginevra Castellano served as the general chair of HRI 2023, with Dr. Katie Winkle as local chair and Dr. Ana Tanevska as online and hybrid chair.	March 13-16, 2023, in Stockholm, Sweden	Researchers and scientific members
NLR				
SISW				